FLAIR Task Schematic

ICU Daily In-Hospital Mortality at 07:00 Local (Task T1)

FLAIR Benchmark

June 4, 2026

Abstract

We specify the FLAIR daily ICU in-hospital mortality task: at every local 07:00 rounds anchor during the first ICU stay, predict whether the hospitalization will end with the patient dying in hospital, defined by CLIF `discharge_category` $\in$ {`Expired`, `Hospice`}. The cohort is all adult ICU hospitalizations — *unconditional*: no `discharge_category` filter, every retained stay contributes daily rows, with null and `Still Admitted` dispositions kept as label-0. Task T2 shares the same cohort and grid but with a different (non-complementary) label.

1 FLAIR operational definition

1.1 Anchors and windows

Let H_i denote a hospitalization with admission time t_i^{adm} and discharge or death time $t_i^{\text{end}} = \min(t_i^{\text{disch}}, t_i^{\text{death}})$. Let $t_i^{\text{ICU}_1,\text{start}}$ and $t_i^{\text{ICU}_1,\text{end}}$ be the first ICU stay endpoints returned by `compute_icu_timing` on the CLIF `ADT` table. All datetimes are naive but wall-clock-local per `cfg["timezone"]`; literal hours of the day are interpreted as local hospital time.

Prediction-window start (pred_start).

$$\text{pred_start}_i \;=\; \text{trunc}(t_i^{\text{ICU}_1,\text{start}},\, 1\text{d}) + 1\text{d} + 7\text{h}.$$

The first decision point is local 07:00 on the calendar day *after* first ICU admission. A patient admitted at 03:00 on Monday gets their first prediction at 07:00 on Tuesday.

Prediction-window stop (pred_stop).

$$\text{pred_stop}_i \;=\; \min(t_i^{\text{ICU}_1,\text{end}},\, \texttt{coalesce}(t_i^{\text{death}},\, t_i^{\text{disch}})).$$

Predictions cease at the end of the first ICU stay or the end of the hospitalization (death if recorded, else discharge), whichever comes first.

Decision-point grid.

Daily 24h step: $D_i = \{\text{pred_start}_i + k{\cdot}24\text{h} : k \in \mathbb{Z}_{\geq 0},\, \text{pred_start}_i + k{\cdot}24\text{h} < \text{pred_stop}_i\}$.

Label (`label_mortality`).

Let $\texttt{disp}_i = \texttt{lower}(\texttt{strip}(\texttt{discharge_category}_i))$. For $d \in D_i$:

$$y_i(d) \;=\; \mathbb{1}[\texttt{disp}_i \in \{\,\texttt{"expired"}, \texttt{"hospice"}\,\}]\,.$$

A null $\texttt{disp}_i$ (i.e., `discharge_category` null in the source) or any non-matching value (Home, SNF, Rehab, LTACH, AMA, Other, `Still Admitted`, ...) yields $y_i(d) = 0$. The label is constant across all rows of H_i (the hospitalization's eventual disposition is a single fact broadcast to every grid point).

1.2 Cohort (unconditional)

- Age at admission ≥ 18 with valid `admission_dttm` and `discharge_dttm`.
- ≥ 1 ICU ADT record (first ICU stay is defined).
- First ICU stay long enough to contain at least one 07:00 grid point after the admission day (i.e., $\text{pred_start}_i < \text{pred_stop}_i$).

There is **no** `discharge_category` filter. Stays discharged Home, SNF, Rehab, LTACH, AMA, Other, with `Still Admitted` status, or with null `discharge_category` all remain in the cohort and carry label-0.

1.3 CLIF tables and categories used

Task T1 reads three CLIF v2.1 tables.

CLIF table	Category column	Usage
`hospitalization`	`discharge_category`	*Label-only*: $y = 1$ iff lower/strip value $\in$ {`Expired`, `Hospice`}; null / `Still Admitted` / any other value $\Rightarrow y = 0$. No cohort filter on this column.
`adt`	`location_category`	Cohort filter: ≥ 1 row with value `"icu"` (case-insensitive).
`patient`	—	join only; no category filter. Provides `death_dttm` for pred_stop.

Table 1: CLIF surface for Task T1. Canonical CLIF `discharge_category` values used by the label (case-insensitive after lower+strip): `Expired`, `Hospice`. All other canonical values (`Home`, `Skilled Nursing Facility (SNF)`, `Acute Inpatient Rehab Facility`, `Long Term Care Hospital (LTACH)`, `Acute Care Hospital`, `Group Home`, `Chemical Dependency`, `Against Medical Advice (AMA)`, `Assisted Living`, `Still Admitted`, `Missing`, `Other`, `Psychiatric Hospital`, `Shelter`, `Jail`) plus null map to label-0.

Columns selected from each table:

- `hospitalization`: `hospitalization_id`, `patient_id`, `admission_dttm`, `discharge_dttm`, `age_at_admission`, `discharge_category`.

- `patient`: `patient_id`, `sex_category`, `race_category`, `ethnicity_category`, `death_dttm`.
- `adt`: `hospitalization_id`, `location_category`, `in_dttm`, `out_dttm`.

2 Algorithmic flow

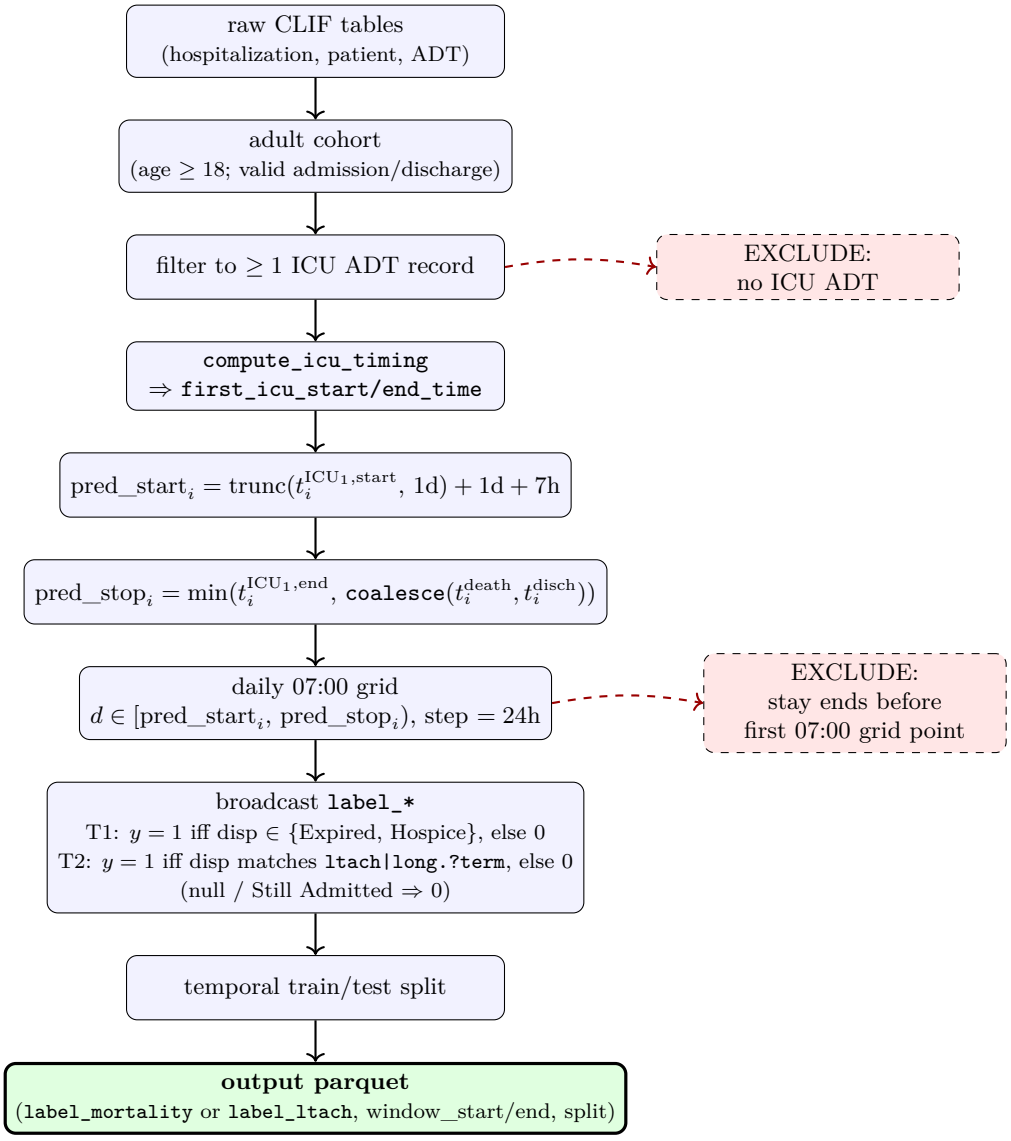


Figure 1: Build pipeline shared by Tasks T1 and T2. The two tasks consume the same cohort, the same grid, and the same lookback window; they differ only in the final label predicate. Task T2 (LTACH) is detailed in a sibling schematic.

Step-by-step prose:

1. Load CLIF tables: `hospitalization`, `patient`, `adt`.
2. Build the adult cohort with valid admission/discharge times.

3. Restrict to hospitalizations with ≥ 1 ICU ADT row.
4. Compute first-ICU-stay window via `compute_icu_timing(adt)`.
5. Compute pred_start$_i$ by truncating $t_i^{\text{ICU}_1,\text{start}}$ to date, adding 1 day, then adding 7 hours.
6. Compute pred_stop$_i = \min(t_i^{\text{ICU}_1,\text{end}}, \texttt{coalesce}(t_i^{\text{death}}, t_i^{\text{disch}}))$.
7. Explode the daily 07:00 grid [pred_start$_i$, pred_stop$_i$) per hospitalization (`pl.datetime_ranges`, `interval = "24h"`, `closed="left"`).
8. Broadcast the label (`disp`$_i \in$ {`"expired"`, `"hospice"`}, null-safe via `fill_null(False)`) to every grid row.
9. Assign `window_start` $= d - 24$h, `window_end` $= d$ (fixed 24h lookback).
10. Apply temporal train/test split per `flair_benchmark._split.assign_split`, keyed on `prediction_dttm`.
11. Select `OUTPUT_COLS` and persist as parquet.

3 Design choices and rationale

Why an unconditional cohort. Earlier drafts of T1 used a conditional-on-non-survival cohort (`discharge_category` $\in$ {LTACH, Expired, Hospice}) to manufacture a balanced binary task. We dropped that design: it conflated the cohort filter with the label predicate, made the per-row positive rate uninterpretable as "ICU mortality risk," and produced a paired-task framing (T1 + T2 = 1) that misrepresented the clinical question. The unconditional design is the standard form of ICU in-hospital mortality prediction (**?**) and makes the per-row positive rate a meaningful ICU mortality prevalence. The task is heavily imbalanced (expected ∼10–15% per-row positive in MIMIC-CLIF); we report AUPRC alongside AUROC.

Why 07:00 local rounds anchor. ICU morning rounds are the canonical decision cadence; predictions produced before rounds are actionable that day, predictions after rounds are not. The 07:00 timestamp is a wall-clock-local literal, relying on `flair_benchmark._clif.pd_to_pl` having stripped timezone info from `clifpy` datetimes (which themselves are typed to `cfg["timezone"]`). Across CLIF sites this is a stable choice; sites that prefer 06:00 or 08:00 can re-anchor by adjusting `PRED_HOUR` without changing the rest of the pipeline.

Why daily (not hourly). The decision being modeled — terminal-trajectory commitment — does not change hour-to-hour. A daily cadence matches the clinical question and keeps row counts manageable.

Why Hospice counts as mortality. CLIF separates `Expired` (death in hospital) from `Hospice` (alive at discharge to hospice services). FLAIR's T1 bundles them: in routine ICU clinical reasoning, transition-to-hospice is the operational equivalent of a death-in-hospital trajectory — goals-of-care have shifted from cure to comfort, and the prediction horizon for the morning-rounds decision is identical. Sites that want the strict CLIF definition (Expired only) can adjust the literal-set in `task1_icu_daily_mortality.py`.

Why null / Still Admitted are kept as label-0. Dropping these would silently shrink the denominator and bias prevalence upward. Treating them as negatives matches the standard in-hospital mortality definition: "did this patient die during this encounter? If we don't see a death-coded disposition, assume no." Sites with persistent `Still Admitted` contamination in their CLIF export should investigate that data-quality gap separately.

4 Out of scope

- Multi-class extension to alive-other dispositions (Home, SNF, Rehab, AMA) or LTACH as a third class.
- Hospital-wide (non-ICU) variant.
- Input feature engineering and severity-score baselines (e.g., APACHE-III (Knaus et al., 1991), SAPS-II).
- Survival-time / time-to-event variant.

References

William A. Knaus, Douglas P. Wagner, Elizabeth A. Draper, Jack E. Zimmerman, Marilyn Bergner, Paulo G. Bastos, Carl A. Sirio, Donald J. Murphy, Ted Lotring, Anne Damiano, and Frank E. Harrell Jr. The APACHE III prognostic system: Risk prediction of hospital mortality for critically ill hospitalized adults. *Chest*, 100(6):1619–1636, 1991. doi: 10.1378/chest.100.6.1619.